

Kinematic Analysis of MW-M31 Simulations: Detailed Methodology

Research Note

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1 Introduction

This note provides a detailed description of a Python-based analysis for computing the kinematic evolution of the Milky Way (MW) and Andromeda (M31) galaxies across simulation snapshots. The main goal is to extract velocity dispersions, rotational support, specific angular momentum, mass distributions, and anisotropy profiles as a function of radius and time.

2 Data Preparation

Simulation snapshots are stored as compressed `.tar` archives, with text files containing particle data. The workflow is as follows:

1. A temporary directory is created to extract snapshot files.
2. All tar files in the current directory are opened.
3. Files corresponding to MW and M31 (with names containing `MW_` or `M31_`) are extracted.

3 Physical Units and Constants

The code adopts the following units and constants:

- Gravitational constant:

$$G = 4.30091 \times 10^{-6} \frac{\text{kpc} \cdot (\text{km/s})^2}{M_{\odot}}$$

where kpc is kiloparsecs, km/s is velocity, and $M_{\odot}$ is solar mass.

- Mass scaling factor: simulation particle masses are multiplied by 10^{10} to convert to solar masses:

$$m_{\text{actual}} = m_{\text{snapshot}} \times 10^{10} M_{\odot}$$

- Radial bins for profile computation are logarithmically spaced between 0.1 kpc and 400 kpc:

$$r_{\text{bins}} = \text{logspace}(-1, 2.6, 20)$$

4 Center of Mass Computation

For each galaxy, the center-of-mass (COM) position and velocity are calculated:

- Particle positions: (x_i, y_i, z_i) .
- Particle velocities: $(v_{x,i}, v_{y,i}, v_{z,i})$.
- Particle masses: m_i .

The COM position is defined as:

$$\mathbf{R}_{\text{COM}} = \frac{\sum_i m_i \mathbf{r}_i}{\sum_i m_i} = (x_{\text{COM}}, y_{\text{COM}}, z_{\text{COM}})$$

The COM velocity is:

$$\mathbf{V}_{\text{COM}} = \frac{\sum_i m_i \mathbf{v}_i}{\sum_i m_i} = (v_{x,\text{COM}}, v_{y,\text{COM}}, v_{z,\text{COM}})$$

If the standard COM velocity computation fails, a fallback uses the mass-weighted average velocity of particles within $r < 15$ kpc from the COM.

5 Relative Velocities and Positions

The position vector relative to the COM for each particle is:

$$\mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{COM}}$$

The velocity relative to the COM is:

$$\mathbf{v}'_i = \mathbf{v}_i - \mathbf{V}_{\text{COM}}$$

6 Radial and Tangential Velocity Components

The radial unit vector is:

$$\hat{\mathbf{r}}_i = \frac{\mathbf{r}'_i}{|\mathbf{r}'_i|}$$

Radial velocity of a particle:

$$v_{r,i} = \mathbf{v}'_i \cdot \hat{\mathbf{r}}_i$$

Tangential velocity magnitude:

$$v_{t,i} = \sqrt{|\mathbf{v}'_i|^2 - v_{r,i}^2}$$

7 Specific Angular Momentum

The specific angular momentum vector for a particle is:

$$\mathbf{j}_i = \mathbf{r}'_i \times \mathbf{v}'_i$$

with magnitude:

$$j_i = |\mathbf{j}_i|$$

The mass-weighted mean specific angular momentum in a radial bin is:

$$j_{\text{spec}} = \frac{\sum_{i \in \text{bin}} m_i j_i}{\sum_{i \in \text{bin}} m_i}$$

8 Radial Profiles and Bin Averaging

Particles are assigned to logarithmic radial bins:

$$\text{bin index } b : r_{\text{bin, low}} \leq r'_i < r_{\text{bin, high}}$$

For each bin, the following quantities are computed:

- Radial velocity dispersion:

$$\sigma_r^2 = \frac{\sum_i m_i (v_{r,i} - \langle v_r \rangle)^2}{\sum_i m_i}, \quad \langle v_r \rangle = \frac{\sum_i m_i v_{r,i}}{\sum_i m_i}$$

- Tangential velocity dispersion:

$$\sigma_t^2 = \frac{\sum_i m_i (v_{t,i} - \langle v_t \rangle)^2}{\sum_i m_i}, \quad \langle v_t \rangle = \frac{\sum_i m_i v_{t,i}}{\sum_i m_i}$$

- Mean rotation velocity about z -axis:

$$v_{\text{rot}} = \left\langle \frac{-x_i v_{y,i} + y_i v_{x,i}}{R_i} \right\rangle, \quad R_i = \sqrt{x_i^2 + y_i^2}$$

- Enclosed mass within radius r :

$$M(< r) = \sum_{i: r'_i \leq r} m_i$$

- Escape velocity at radius r :

$$v_{\text{esc}} = \sqrt{\frac{2GM(< r)}{r}}$$

- Velocity anisotropy:

$$\beta = 1 - \frac{\sigma_t^2}{2\sigma_r^2}$$

9 Time Evolution

The code processes multiple snapshots to produce time series of all quantities. Global averages over the inner 30 kpc are computed:

$$\langle \sigma_r \rangle_{\text{inner}} = \text{mean of } \sigma_r \text{ for } r \leq 30 \text{ kpc,} \quad \text{similarly for } \sigma_t, v_{\text{rot}}, j_{\text{spec}}$$

10 Visualization

The code produces:

- Line plots of inner kinematics versus time.
- Heatmaps of $\log_{10}(\sigma_r)$ and β as a function of radius and snapshot.

11 Conclusion

This pipeline allows for a systematic computation of galaxy kinematics and dynamical properties across time. The analysis provides:

- Insights into radial and tangential velocity dispersions.
- Evolution of rotational support and angular momentum.
- Mass distribution and escape velocity profiles.
- Anisotropy structure as a function of radius.